DISSECTING CHRONOS: SPARSE AUTOENCODERS REVEAL CAUSAL FEATURE HIERARCHIES IN TIME SERIES FOUNDATION MODELS

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710M Chronos-T5-Large parameters

49,152 SAE features across 6 layers

392 single-feature ablations

100% positive ΔCRPS

38.61 max single-feature ΔCRPS

Main takeaway. Sparse autoencoders show that Chronos-T5-Large organizes temporal concepts by depth: early encoder layers encode local frequency and volatility, the mid-encoder is the main causal bottleneck, and the final encoder is semantically richest but less critical on ETT.

Why this question matters

Time series foundation models are increasingly used in high-stakes forecasting, but their internal representations remain opaque. This work is the first application of sparse autoencoders to a time series foundation model and asks:

- Can SAE features in Chronos-T5 be interpreted as temporal concepts?
- Which layers contain the features that the model causally depends on for forecasting?
- Does semantic richness align with causal importance?

Method at a glance

Chronos-T5-Large activations → Train TopK SAEs → Synthetic diagnostics → Single-feature ablation → Progressive ablation + CRPS

Base model	Chronos-T5-Large, 710M parameters, encoder-decoder T5 architecture, 4,096-bin tokenization
Extraction points	Encoder blocks 5, 11, and 23; decoder blocks 11 and 23; cross-attention output at decoder block 11
SAE setup	$d_{\text{model}} = 1024$, $d_{\text{SAE}} = 8192$, TopK $= 64$, 50,000 training steps, MSE reconstruction
Data	Synthetic diagnostics for labeling and ETT for causal evaluation

SAE update used in the paper

$$z = \mathrm{TopK}(W_{\text{enc}}(x - b_{\text{dec}}) + b_{\text{enc}},\ k), \qquad \hat{x} = W_{\text{dec}} z + b_{\text{dec}}.$$

Depth-dependent temporal hierarchy

The taxonomy reveals a clear layer progression. Semantic coverage rises from 4.9% in the early encoder to 25.8% in the mid-encoder and 59.8% in the final encoder, while decoder layers remain sparsely labeled.

Early encoder (block 5) — **4.9% labeled**

Dominated by **high frequency** and **high volatility** features, suggesting local feature extraction.

Mid-encoder (block 11) — **25.8% labeled**

Dominated by **level shift up** (1,024 features), **noise** (413), and **high volatility** (268), with seasonality nearly absent. This layer acts as a change-detection hub.

Final encoder (block 23) — **59.8% labeled**

Richest semantic layer, with strong **seasonality**, **level shift up**, and **high/low frequency** coverage. It compresses the broadest taxonomy before cross-attention.

Decoder layers — **3.1–5.5% labeled**

Low labeling overall, with **low volatility** and **low frequency** relatively prominent, consistent with smooth forecast generation.

What the paper contributes

- First application of sparse autoencoders to a time series foundation model.
- 392/392 encoder feature ablations increase CRPS, showing universal causal relevance.
- Causal importance and semantic richness diverge: the **mid-encoder** is most critical, while the **final encoder** is most semantically diverse.

Selected references

Ansari et al. *Chronos: Learning the Language of Time Series.* TMLR, 2024.
Cunningham et al. *Sparse Autoencoders Find Highly Interpretable Features in Language Models.* ICLR, 2024.
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Jin et al. *Time-LLM: Time Series Forecasting by Reprogramming Large Language Models.* ICLR, 2024.

Single-feature ablations: every tested feature matters

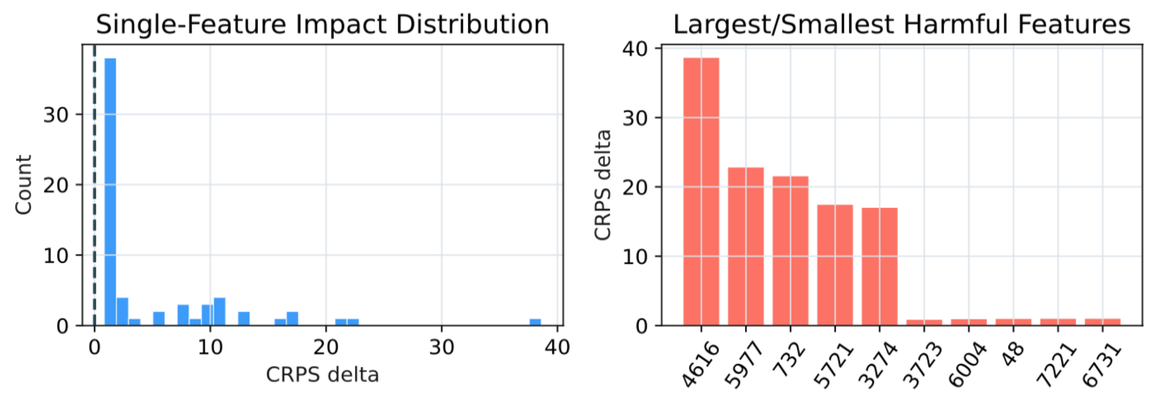

Figure. Distribution of single-feature CRPS deltas at encoder block 11 and the most/least harmful individual features. All tested feature ablations produce positive ΔCRPS.

Layer	n	Mean	Med.	Max	+Frac
Enc. 5	64	3.05	0.95	26.32	1.00
Enc. 11	64	5.15	1.26	38.61	1.00
Enc. 23	64	3.73	2.98	11.65	1.00
Enc. 23†	200	2.37	2.37	2.44	1.00

†Extended sweep uses 1,024 windows and 8 forecast samples. The mid-encoder has the heaviest right tail, with the most extreme single-feature dependence.

Progressive ablation reveals the causal bottleneck

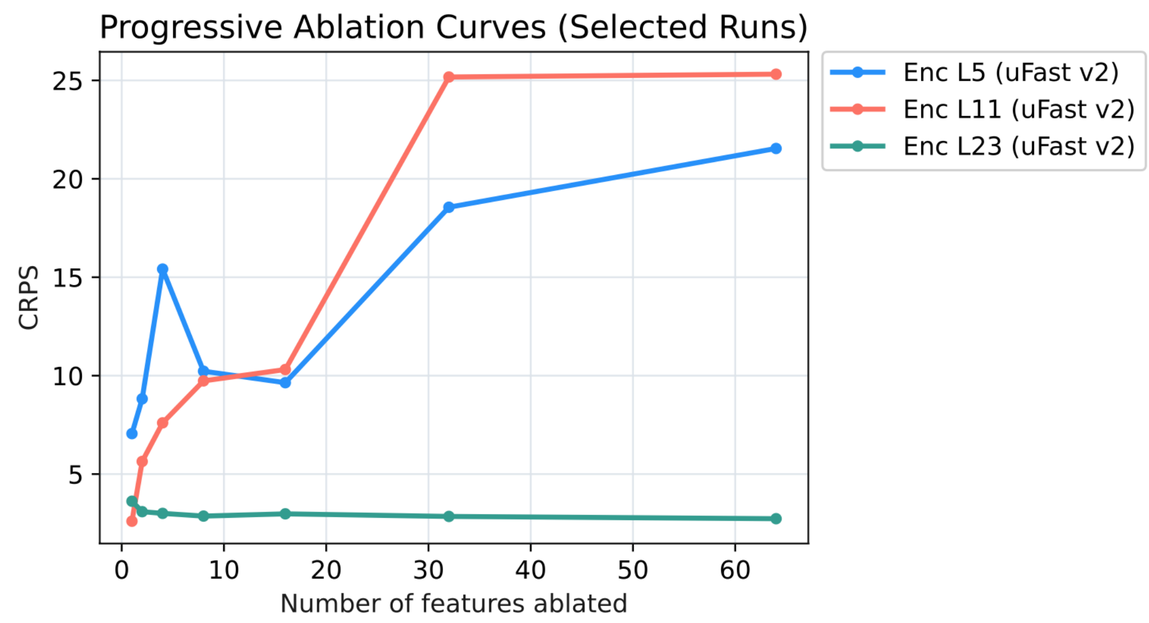

Figure. Cumulative ablation of top-ranked features. Early and mid-encoder layers degrade sharply, while the final encoder remains flat or slightly improves.

Encoder block 11
CRPS rises from **2.61** to **25.32** after 64 top-feature ablations. This is the model's strongest causal bottleneck.

Encoder block 5
CRPS rises from **7.05** to **21.54**. Early layers matter, but less than block 11.

Encoder block 23
CRPS improves from **3.62** to **2.73**; the 200-feature run stays nearly flat at **3.59** to **3.58**.

Interpretation. On ETT, the semantically richest layer is *not* the most causally essential one. Chronos-T5 relies primarily on **abrupt-dynamics detection** rather than periodic-pattern encoding.

Key numerical story for the poster

- **Universal causal relevance:** 392/392 single-feature ablations produce positive ΔCRPS.
- **Heavy-tailed importance:** encoder block 11 reaches max/median = 30.5×, versus 3.9× at encoder block 23.
- **Inverse relationship:** the final encoder has the richest taxonomy but lower and more uniform per-feature impact.

Discussion, limitations, and conclusion

Main conclusion

Sparse autoencoders transfer effectively to time series foundation models. They expose meaningful temporal features in Chronos-T5-Large and identify the mid-encoder as the main computational bottleneck for ETT forecasting.

Why the paradox matters

The final encoder may contain features that help broad pretraining-domain generalization but are suboptimal for ETT. Progressive ablation therefore behaves like mild implicit domain adaptation.

Limits of the current study

The taxonomy is heuristic and leaves 82.8% of features unlabeled overall. Causal experiments are run only on ETT and only for Chronos-T5-Large. The ultra-fast ablation setting is directionally useful but statistically limited.